

Accessible Documents in LaTeX

https://github.com/rhstanton/accessible_LaTeX

Version 1.4

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1 Introduction

On January 8, 2026, we were notified by campus that, beginning in April 2026:

“The updated requirements of the ADA require that digital course materials provided to students, even materials inside password-protected course sites like bCourses, will need to comply with accessibility standards (Web Content Accessibility Guidelines (WCAG) 2.1 Level AA).”

Many of us use L^AT_EX to create teaching materials—both slides and documents. Standard L^AT_EX (including Beamer) does not automatically generate accessible PDFs.

2 What is this project?

An experiment exploring accessible L^AT_EX documents using the L^AT_EX Tagging Project:

- **Two templates:** Articles (`article` class) and slides (`ltx-talk`)
- **Shows:** What’s common to both, what’s different
- **Contains:** Working examples with math, text, graphics, and tables
- **Scores:** Perfect 100% from the bCourses accessibility checker (Ally)

2.1 How to use it

1. **Learn the common requirements:** Apply to any L^AT_EX document
2. **See class-specific needs:** Understand slides vs. articles
3. **Copy and adapt:** Use as templates for your documents
4. **Study the code:** Heavily commented for learning

Available at: https://github.com/rhstanton/accessible_LaTeX

2.2 Converting to accessible LaTeX: Overview

- **Your existing L^AT_EX skills transfer** - you're just adding accessibility features
- **The changes are minimal** and follow consistent patterns

Common requirements (both slides & articles):

- Add document metadata and enable PDF tagging
- Tag images with alt text; tag table headers
- Use accessible colors; compile with LuaLaTeX

Key difference:

- Articles: **Keep your existing class!** Just add accessibility features
- Slides: Switch to `ltx-talk` (similar syntax, but recreate styling)

Both require LaTeX kernel 2025-11-01 (see next sections for details).

See “Getting started” sections at the end for detailed steps. For complete LaTeX Tagging Project documentation, see [latest detailed instructions](#).

3 TeX version requirements (both slides and articles)

- **Minimum:** TeX Live 2023 or later with all packages updated
- **Critical:** Must have LaTeX kernel 2025-11-01
 - Update via TeX Live Manager (Windows) or TeX Live Utility (Mac)
 - Or use Overleaf Labs (see next section)
- **Will NOT work:** TeX Live 2022 or earlier

4 You CAN use Overleaf (both slides and articles)

- `ltx-talk` requires a very recent TeX Live version
- This is available through Overleaf's [Labs program](#) (not standard Overleaf)

4.1 Using Overleaf (2 steps)

1. Join Overleaf Labs:

- Visit <https://www.overleaf.com/labs/participate>
- Opt in and enable “Rolling TeX Live releases”

2. Configure project:

- Set TeX Live version to “**Rolling TeXLive (labs)**” (bottom of list)
- Set Compiler to **LuaLaTeX**

Resources: <https://docs.overleaf.com/writing-and-editing/creating-accessible-pdfs>

5 Common requirements for ALL documents

Whether you’re creating slides or articles, these requirements are [the same](#):

5.1 Document Metadata

- Every accessible document **must** begin with `\DocumentMetadata`
- Goes *before* `\documentclass`

```
\DocumentMetadata{
  pdfstandard=a-2u,      % PDF/A-2u format
  pdfstandard=ua-1,      % Add PDF/UA-1 conformance target
  pdfversion=1.7,        % Explicit PDF version
  lang=en-US,            % Language for screen readers
  tagging=on,            % Enable PDF tagging
  tagging-setup={
    math/alt/use,         % Keep math accessibility enabled (screen readers get tagged math)
    math/mathml/AF=false, % Do not embed MathML as PDF Associated Files
    math/tex/AF=false,    % Do not embed TeX source as Associated Files
    math/mathml/sources= % Clear MathML source attachment list
  }
}
```

This configures accessibility, math tagging, and validator-compatibility behavior in one place. The `tagging-setup` entries keep math accessible while reducing helper attachments that can trigger warnings in some strict conformance validators.

5.2 Images, Tables, and Colors

- **Images:** All images must include alt text descriptions
 - Allows screen readers to describe visual content
 - Even decorative images need marking
 - See dedicated “Figures” section for examples
- **Tables:** Must specify which rows are headers
 - Helps screen readers navigate table structure
 - Essential for data accessibility
 - See dedicated “Tables” section for examples
- **Accessible colors:** WCAG 2.1 requires 4.5:1 contrast ratio

```
\colorlet{AccessibleRed}{red!80!black}
\colorlet{AccessibleGreen}{green!40!black}
% Standard blue is fine as-is
```

5.3 LuaLaTeX Compilation

- You **must** compile with LuaLaTeX (not pdfLaTeX or XeLaTeX)
- **Why LuaLaTeX?**
 1. **Automatic MathML:** Makes math accessible without manual work
 2. **Full tagging support:** Complete PDF accessibility features
 3. **Modern fonts:** Handles Unicode and OpenType fonts
- **How to switch:**
 - Command line: `lualatex filename.tex`
 - Most editors: Select “LuaLaTeX” from compiler menu

5.4 Optional validator cleanup for some checkers (`strip_af.py`)

- Even with the metadata settings above, some builds may still include PDF Associated File entries.
- These are typically harmless helper attachments, but some strict conformance checkers can still flag them.
- Ally can still score 100% without this step; run `strip_af.py` when you need checker-specific conformance cleanup.

```
python build/strip_af.py accessible_article.pdf
```

This writes `accessible_article.noaf.pdf`, removing /AF and embedded-file structures while preserving document content and accessibility tagging.

6 The basics

- Use standard L^AT_EX environments: `section`, `subsection`, `itemize`, `enumerate`, etc.
- **Existing source files don’t need a lot of editing**
- Here’s some gratuitous *math* for the accessibility checker

7 Figures

When including a figure, you **must provide alt text** describing the image for screen readers. This is *required* for accessibility compliance.

```
\includegraphics[height=.4\textheight,alt={A capybara}]{capybara.jpg}
```



Figure 1: A picture of a capybara

8 Tables

When including a table, you **must specify which rows are headers**. Screen readers use this information to help users navigate table content. Use `\tagpdfsetup{table/header-rows={...}}` *before* the `tabular` environment:

- Use `{1}` for 1 header row
- Use `{1,2}` for 2 header rows
- Use `{1,2,3}` for 3 header rows, etc.
- **Validator note:** PAC can sometimes report `Table header cell has no associated subcells` even when headers are tagged correctly. The LaTeX team documents this behavior at <https://github.com/latex3/tagging>

8.1 Example: Table with 3 Header Rows

```
\tagpdfsetup{table/header-rows={1,2,3}}
\begin{tabular}{cccccccr}
\toprule
← 3 header rows
\midrule
← data rows
\bottomrule
\end{tabular}
```

Payment date	Caplet expiry date	DF_{pay}	Forward rate	Days to expiry	Days in accrual period	T_{expiry}	Δ	Caplet
2004/12/01	—	0.99550	0.01790	0	91	0.00000	0.25278	—
2005/03/01	2004/11/29	0.99008	0.02188	89	90	0.24384	0.25000	1,178.77
2005/06/01	2005/02/25	0.98401	0.02413	177	92	0.48493	0.25556	4,844.73
2005/09/01	2005/05/27	0.97733	0.02675	268	92	0.73425	0.25556	10,016.71

Table 1: A table

9 Common pitfalls

- **Forgetting alt text for images**
 - Every `\includegraphics` needs an `alt={...}` parameter
 - Even decorative images need alt text (use `alt={decorative}`)
- **Not specifying table header rows**
 - Add `\tagpdfsetup{table/header-rows={...}}` before each table
 - Use `{1}` for 1 header row, `{1,2}` for 2 header rows, etc.
- **Insufficient color contrast**
 - WCAG 2.1 requires 4.5:1 contrast ratio for normal text
 - Avoid light colors: `yellow`, `cyan` fail contrast requirements
 - Darken `red` and `green`: use `red!80!black`, `green!40!black`
 - Standard `blue` is fine and meets WCAG requirements
 - Test with a contrast checker: <https://webaim.org/resources/contrastchecker/>
- **Using the wrong compiler**
 - Make sure your editor is set to use `LuaLaTeX`, not `pdfLaTeX`
- **Old TeX distribution**
 - TeX Live 2022 or earlier won't work
 - Update packages using TeX Live Manager (Windows) or TeX Live Utility (Mac)

10 Validator-specific notes

Not all validator warnings indicate real problems. Some are *validator bugs*:

- **Acrobat: “Lbl and LBody – Failed”**
 - Appears around figure captions, section numbers, table captions
 - **This is a validator bug**, not a source error
 - The `<Lbl>` tag is correctly used for numbering
 - Acrobat's checker incorrectly expects it only in lists
 - **Safe to ignore** – your PDF structure is correct
 - References:
 - * LaTeX developer Ulrike Fischer: “Adobe doesn't like the (valid) use of Lbl for the section numbers, complain to them that they should correct their checker.” (<https://tex.stackexchange.com/a/709655/238>)
 - * Adobe Community expert Philip Kiff: “`<Lbl>` is getting flagged as an error by Acrobat Pro DC by mistake... you could just ignore this error and wait until Acrobat fixes their built-in tester.”
- **PAC: “Table header cell has no associated subcells”**
 - Often a validator interpretation issue
 - Verify your `\tagpdfsetup{table/header-rows={...}}` is correct
 - PAC may not fully handle attribute-class form for table scope
 - Known behavior: <https://github.com/latex3/tagging-project/issues/1056>

Recommendation: Use multiple validators (Ally, veraPDF, PAC) and compare findings. No single checker catches everything, and each may have quirks.

11 Getting started: Common steps for both

1. **Setup environment:** Use Overleaf Labs OR install/update TeX Live locally
 - Overleaf: See earlier section for Labs setup
 - Local: Update via TeX Live Manager (Windows) or TeX Live Utility (Mac)
2. **Get templates:** Download from https://github.com/rhstanton/accessible_LaTeX
3. **Add accessibility features:**
 - Add `alt` text to images
 - Add `table/header-rows` to tables
4. **Set compiler to LuaLaTeX**
5. **Compile:**
 - Compile with LuaLaTeX

12 Getting started: What's different by document type

12.1 For Articles (using `article`, `report`, `book`, etc.)

- Add `\DocumentMetadata` before `\documentclass`
- Include the `tagging-setup` block in `\DocumentMetadata` (as shown in this template)
- Switch to `fontspec` and `unicode-math` packages
- **Keep your existing `documentclass`!**

12.2 For Slides (migrating from Beamer)

- Copy preamble from `accessible_slides.tex`
- Change `\documentclass{beamer}` to `\documentclass[frame-title-arg]{ltx-talk}`
- **Remove Beamer themes/templates/colors**
- **Recreate styling** using standard LaTeX/xcolor
- **One-time preamble work**—then reuse for all future talks!

13 Validation workflow: use multiple checkers

- No single checker catches everything.
- Accessibility usability checks and strict PDF conformance checks are not identical.
- A practical workflow is to run more than one checker and compare findings.

Checkers used so far: Ally (the accessibility checker built into bCourses), veraPDF (open-source PDF conformance validator, <https://verapdf.org/>), and PAC (PDF Accessibility Checker, <https://pac.pdf-accessibility.org/>).

14 Optional validator cleanup step: `strip_af.py`

- The metadata settings alone are enough to generate a 100% score on Ally.
- Stricter checkers (e.g., **veraPDF**) may still complain about /AF or embedded-file structures.
- Optional strict-archival cleanup command:

```
python strip_af.py build/yourfile.pdf build/yourfile.noaf.pdf
```

- Validate `build/yourfile.noaf.pdf` with veraPDF.
- Prerequisite for the script: `python -m pip install pikepdf`

15 Validator-specific notes

- This is often a validator interpretation issue, not necessarily a source-tagging error.
- PAC table warning (known behavior)
 - PAC can report `Table header cell has no associated subcells` even when `table/header-rows` is correct
 - The LaTeX Tagging Project documents this behavior at <https://github.com/latex3/tagging-project/issues/1056>
- When this warning appears
 - First confirm source markup is correct (especially `table/header-rows`)
 - Keep source tagging and document the validator warning
 - Report minimal reproductions to tool vendors when possible
- **Interpretation tip:** Treat one-tool warnings as signals to investigate, not automatic proof of source errors.

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